

Time Series Analysis

Seminar 7: Cointegration and VECM



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Acronyms used in this material

- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model)
- **BIC**: Bayesian Information Criterion
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- **ECM**: Error Correction Model
- **EODHD**: EOD Historical Data (market-data provider)
- **FRED**: Federal Reserve Economic Data
- **GDP**: Gross Domestic Product
- **HUF**: Hungarian forint (ISO 4217 code)
- **OLS**: Ordinary Least Squares
- **PLN**: Polish zloty (ISO 4217 code)
- **ROBOR**: Romanian Interbank Offer Rate
- **SE**: Standard Error
- **US**: United States
- **VAR**: Vector AutoRegression
- **VECM**: Vector Error Correction Model

Today's Question and Seminar Structure

- The question of the seminar
  - **are two or more $I(1)$ series tied by a long-run equilibrium, how fast do they return to it, and who does the adjusting?**
- Results to obtain (practice only: nothing is handed in or graded)
  - the answers to the A exercises
  - a figure and a table for each B exercise
  - the corrected AI answer of C2
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- The exercises, their types and their order: the exercise map (next slide)
- Seminar: prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	an Engle–Granger statistic with two and with three variables	Solved, Proposed	A1
A3, A4	an ECM and a bivariate VECM: speed and half-life	Solved, Proposed	A3
A5, A6	Johansen statistics from eigenvalues; a VAR(2) written as a VECM	Solved, Proposed	A5
B1, B2	Engle–Granger and ECM: US yields; Romanian consumption and GDP	Solved, Proposed	B1
B3, B4	Johansen and VECM: three currencies; three interest rates	Solved, Proposed	B3
C1, C2	pairs trading out of sample; what is wrong in an AI answer?	Proposed	B1, B3

- Types: **A** calculations on paper; **B** real data, in the notebook; **C** open questions, an AI critique
- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in the seminar

Data used

Series	Source	Frequency	Period
US Treasury yields, 1 and 10 years	FRED (GS1, GS10)	monthly	1960–2026
Household consumption and GDP, Romania	Eurostat (namq_10_gdp)	quarterly, SA	1995–2026
EUR/RON, EUR/HUF, EUR/PLN	BNR reference rates	month-end	2005–2026
ROBOR 3M, Euribor 3M	Eurostat (irt_st_m)	monthly	2010–2026
Romanian 10-year yield	EODHD	daily, monthly mean	2010–2026
Banca Transilvania, BRD	EODHD	daily, adjusted close	2014–2026

- Levels as $100 \ln$ (consumption, GDP, exchange rates, share prices); interest rates in % per year
- In the notebook: `adf_test`, `eg_test` (Engle–Granger and Phillips–Ouliaris), `ecm_fit`, `johansen`; no account or key is needed

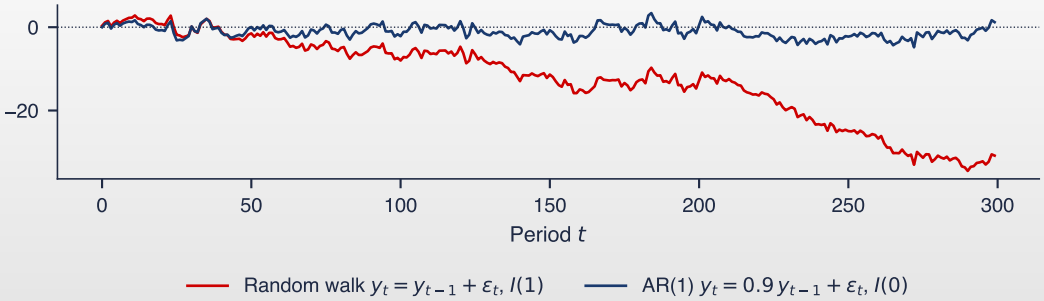
Prerequisites for Today (1/15): unit roots and the ADF test

- $y_t \sim I(1)$: y_t has a unit root and its difference $\Delta y_t = y_t - y_{t-1}$ is stationary, $I(0)$ (Chapter 3)
  - $I(d)$: stationary after d differences; a shock to an $I(1)$ series never dies out
- **ADF test** (augmented Dickey–Fuller): a regression of the change on the lagged level

$$\Delta y_t = c\ [+\ \delta t] + \gamma\, y_{t-1} + \sum_{j=1}^{k} \delta_j \Delta y_{t-j} + e_t, \qquad \tau = \frac{\hat{\gamma}}{\mathrm{SE}(\hat{\gamma})}$$

  - c: a constant; δt: an optional linear trend; δ_j: the coefficients of the k lagged differences, which make e_t white noise
  - $\gamma = \phi - 1$ for an AR(1) $y_t = \phi y_{t-1} + \ldots$; H_0: $\gamma = 0$ (unit root); H_1: $\gamma < 0$ (stationary)
  - SE: the standard error; reject H_0 if τ is below the Dickey–Fuller critical value (constant, 5%: -2.86), not -1.645

Prerequisites for Today (2/15): a random walk and a stationary AR(1)



- The same shocks ε_t; only the coefficient of y_{t-1} differs: 1 or 0.9
- AR(1): returns to its mean 0, stays within about ± 5
  - random walk: wanders away (down to -35) and has no mean to return to; its variance grows as $t\sigma^2$

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Prerequisites for Today (3/15): the spurious regression

- **Spurious regression** Granger and Newbold (1974): OLS of one $I(1)$ series on another, unrelated one

$$y_t = a + b\,x_t + u_t, \qquad x_t, y_t \text{ independent random walks}$$

  - u_t is itself $I(1)$: the usual t and F tests are invalid, $|t| > 1.96$ far more often than 5%
  - typical signs: a high R^2 and strongly autocorrelated residuals
- **Correlation in levels** of $I(1)$ series proves nothing
  - compare the correlation of the changes Δx_t, Δy_t, which are stationary
  - the exception: cointegrated series, for which u_t is stationary (next slides)

Prerequisites for Today (4/15): independent random walks and the t test



- Left: two independent random walks, $T = 200$; they seem to move together, then apart
- Right: the t statistic of $\hat{b}$ in 2 000 such regressions, against the $N(0, 1)$ reference
  - $|t| > 1.96$ in 82% of the pairs instead of 5%; median R^2: 0.17, with no relation at all

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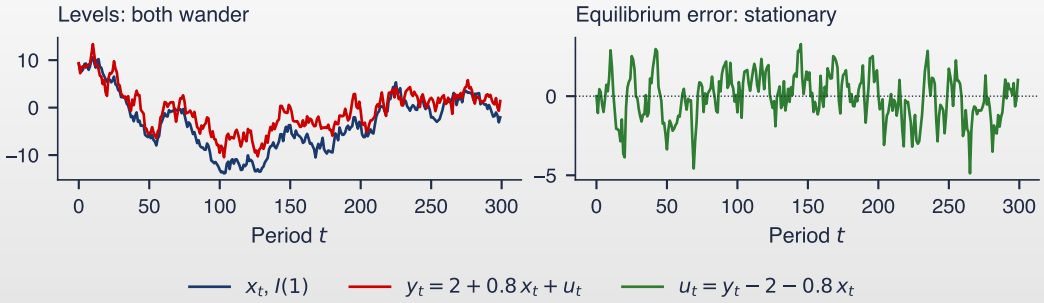
Prerequisites for Today (5/15): cointegration

- **Cointegration** Engle and Granger (1987): $I(1)$ series with a stationary linear combination

$$\mathbf{y}_t = (y_{1t}, \ldots, y_{nt})^\top \sim I(1), \qquad u_t = \boldsymbol{\beta}^\top \mathbf{y}_t - \mu \sim I(0)$$

  - $\mathbf{y}_t$: the vector of the n series; $^\top$: the transpose; $\boldsymbol{\beta}$: the cointegrating vector, normalised with one coefficient equal to 1
  - u_t: the equilibrium error, μ: its mean; two series: $y_{1t} = \mu + b\, y_{2t} + u_t$, $\boldsymbol{\beta} = (1, -b)^\top$
- **Common trends** Stock and Watson (1988): n variables, r cointegrating vectors, $n - r$ common stochastic trends
  - example: the 1-year and the 10-year interest rate wander, but their spread stays within a band
  - a known $\boldsymbol{\beta}$, such as $(1, -1)$ for $c_t - y_t$ (log consumption minus log GDP, a "great ratio"): test ADF directly on $\boldsymbol{\beta}^\top \mathbf{y}_t$

Prerequisites for Today (6/15): a cointegrated pair



- Simulated: x_t a random walk, $y_t = 2 + 0.8\,x_t + u_t$ with $u_t = 0.7\,u_{t-1} + e_t$ stationary
- Left: both levels wander between about -14 and 11, together
  - right: the equilibrium error u_t returns to 0 (standard deviation 1.5); $\beta = (1, -0.8)^\top$, one common trend

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Prerequisites for Today (7/15): the Engle–Granger test

- **Step 1**: OLS in levels, then the residuals

$$y_t = a + b\,x_t + u_t, \qquad \hat{u}_t = y_t - \hat{a} - \hat{b}\,x_t$$

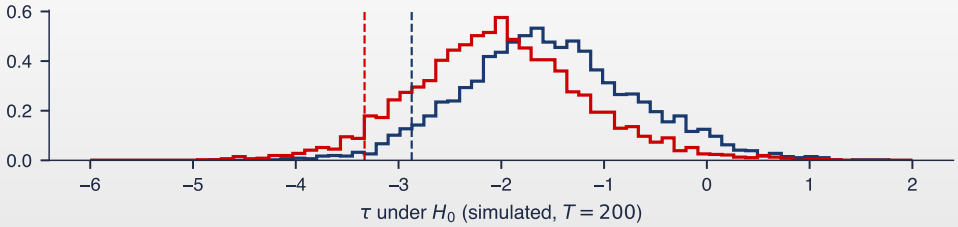
- **Step 2**: an ADF test on $\hat{u}_t$, without constant ($\hat{u}_t$ has mean 0)

$$\Delta\hat{u}_t = \gamma\,\hat{u}_{t-1} + \sum_{j=1}^{k} \delta_j \Delta\hat{u}_{t-j} + e_t, \qquad \tau = \frac{\hat{\gamma}}{\mathrm{SE}(\hat{\gamma})}$$

  - H_0: no cointegration ($\gamma = 0$, $\hat{u}_t$ has a unit root); reject if τ is below the critical value

- **Critical values** MacKinnon (1994), constant, large T, 5%: more negative than Dickey–Fuller
  - -2.86 (one series, Dickey–Fuller); -3.34 (2 variables); -3.74 (3 variables)
  - **Phillips–Ouliaris** Z_t Phillips and Ouliaris (1990): the same H_0 and critical values; no lags, a long-run variance correction instead

Prerequisites for Today (8/15): why the critical values are more negative



- Simulated τ under H_0, $T = 200$, 4 000 replications: blue for one random walk (constant), red for the residuals of two independent random walks
- 5% quantiles: -2.87 and -3.33, close to -2.86 and -3.34
  - OLS chooses $\hat{b}$ to minimise the variance of $\hat{u}_t$, so $\hat{u}_t$ looks more stationary than it is

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Prerequisites for Today (9/15): the error correction model

- **ECM** (error correction model) Davidson et al. (1978): the change of y reacts to last period's distance from equilibrium

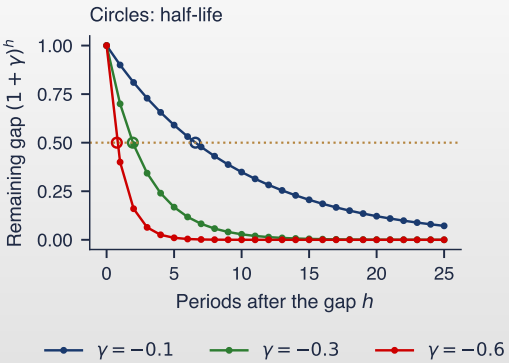
$$\Delta y_t = c + \gamma\,\hat{u}_{t-1} + \delta_0\,\Delta x_t + \cdots + \varepsilon_t, \qquad \hat{u}_{t-1} = y_{t-1} - \hat{a} - \hat{b}\,x_{t-1}$$

  - $\gamma < 0$: the **speed of adjustment**, the share of last period's gap closed by y; $\gamma \geq 0$: no error correction
  - δ_0: the short-run effect of x; b: the long-run effect; c: the constant; "…": lags of Δy and Δx; ε_t: white noise
  - all the terms are stationary: OLS and t-tests are valid
- **Half-life**: without new shocks a gap shrinks like $(1+\gamma)^h$

$$(1+\gamma)^{h_{1/2}} = 0.5 \iff h_{1/2} = \frac{\ln 0.5}{\ln(1+\gamma)}$$

  - example: $\gamma = -0.5$ gives $h_{1/2} = 1$; $\gamma = -0.1$ gives about 6.6 periods

Prerequisites for Today (10/15): how fast a gap closes



- A gap of 1 at $h = 0$
  - remaining after h periods: $(1+\gamma)^h$
- Half-lives (circles)
  - $\gamma = -0.1$: 6.6 periods
  - $\gamma = -0.3$: 1.9 periods
  - $\gamma = -0.6$: 0.8 periods
- With monthly data, $\gamma = -0.1$
  - means half a year to close half of a gap

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Prerequisites for Today (11/15): the VECM

- **VECM** (vector error correction model): a VAR(p) of Chapter 6 rewritten in differences

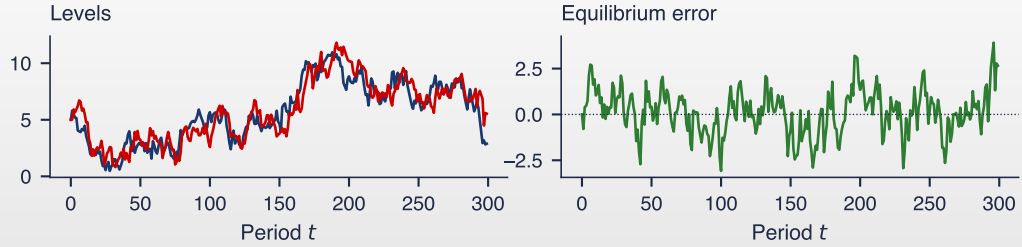
$$\Delta \mathbf{y}_t = \mathbf{\Pi}\, \mathbf{y}_{t-1} + \sum_{i=1}^{p-1} \mathbf{\Gamma}_i \Delta \mathbf{y}_{t-i} + \mathbf{u}_t, \qquad \mathbf{\Pi} = \mathbf{A}_1 + \cdots + \mathbf{A}_p - \mathbf{I}, \quad \mathbf{\Gamma}_i = -(\mathbf{A}_{i+1} + \cdots + \mathbf{A}_p)$$

  - from $\mathbf{y}_t = \mathbf{A}_1 \mathbf{y}_{t-1} + \cdots + \mathbf{A}_p \mathbf{y}_{t-p} + \mathbf{u}_t$; $\mathbf{A}_i$: the $n \times n$ VAR matrices; $\mathbf{I}$: the identity; $\mathbf{u}_t$: white-noise shocks
- $r = \text{rank}(\mathbf{\Pi})$: the number of cointegrating vectors

$$0 < r < n: \quad \mathbf{\Pi} = \boldsymbol{\alpha}\boldsymbol{\beta}^\top, \qquad \boldsymbol{\alpha}, \boldsymbol{\beta} \text{ of size } n \times r$$

  - $r = 0$: no cointegration (a VAR in differences); $r = n$: all series stationary
  - columns of $\boldsymbol{\beta}$: the cointegrating vectors; $\boldsymbol{\alpha}$: the **adjustment coefficients** (loadings) of each variable to each equilibrium error
  - $\alpha_i = 0$: y_i does not adjust, it is **weakly exogenous**
  - **Granger representation theorem** Engle and Granger (1987): with cointegration at least one $\alpha_i \neq 0$

Prerequisites for Today (12/15): a VECM with one adjusting variable



- $\Delta y_{1t} = \alpha_1 z_{t-1} + u_{1t}$, $\Delta y_{2t} = \alpha_2 z_{t-1} + u_{2t}$, with $z_t = y_{1t} - y_{2t}$, $\boldsymbol{\beta} = (1, -1)^\top$
- $\alpha_1 = -0.2$, $\alpha_2 = 0$: y_2 is a random walk (weakly exogenous), y_1 follows it
  - the levels share one common trend, the trend of y_2; z_t stays around 0

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Prerequisites for Today (13/15): the Johansen tests

- **Johansen** Johansen (1988): maximum likelihood for the VECM gives eigenvalues $1 > \hat{\lambda}_1 \geq \cdots \geq \hat{\lambda}_n \geq 0$
  - $\hat{\lambda}_i$: the squared canonical correlation between $\Delta \mathbf{y}_t$ and a combination of $\mathbf{y}_{t-1}$; large: a stationary combination; ≈ 0: still $I(1)$
- Two statistics, with T the number of observations

$$\lambda_{\text{trace}}(r) = -T \sum_{i=r+1}^{n} \ln(1 - \hat{\lambda}_i), \qquad \lambda_{\max}(r) = -T \ln(1 - \hat{\lambda}_{r+1})$$

  - trace: H_0: rank $\leq r$ against rank $= n$; maximum eigenvalue: H_0: rank $= r$ against rank $= r + 1$
  - large statistics (eigenvalues far from 0): reject H_0; non-standard critical values Osterwald-Lenum (1992)
- **Sequential procedure**: test $r = 0$, $r \leq 1$, ...; the rank is the first r not rejected
  - 5% critical values with a constant, $n = 3$: trace 29.80, 15.49, 3.84; max 21.13, 14.26, 3.84
  - **restricted constant**: the constant enters only $\boldsymbol{\beta}^\top \mathbf{y}_{t-1} + \mu$, the levels have no drift

Prerequisites for Today (14/15): pairs trading

- **Spread** of two cointegrated prices Gatev, Goetzmann and Rouwenhorst (2006), standardised with formation-period moments

$$s_t = \ln P_t^A - \hat{b} \ln P_t^B, \qquad z_t = \frac{s_t - \bar{s}}{\hat{\sigma}_s}$$

  - P_t^A, P_t^B: the two prices; $\hat{b}$: the Engle–Granger slope; $\bar{s}$, $\hat{\sigma}_s$: mean and standard deviation of s_t in the formation period

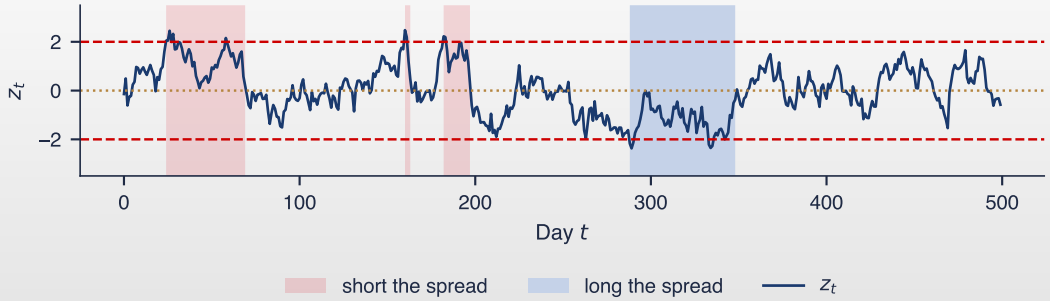
- **Rule**: open at $|z_t| > 2$, close when z_t returns to 0
  - $z_t > 2$: sell A, buy $\hat{b}$ units of B (short the spread); $z_t < -2$: the reverse (long the spread)
  - parameters fixed in the formation period, applied in a later trading period: an out-of-sample test

- **Sharpe ratio**: return per unit of risk

$$\mathrm{SR} = \frac{\bar{r}_{\text{annual}}}{\sigma_{\text{annual}}}$$

  - $\bar{r}_{\text{annual}}$: the mean annual return of the strategy; σ_{annual}: its annual standard deviation; net: after transaction costs

Prerequisites for Today (15/15): the z-score rule



- A simulated stationary spread ($s_t = 0.95\, s_{t-1} + e_t$), 500 days, standardised
- 4 trades: each opens at ± 2 (dashed) and closes at 0 (dotted); 25% of the days in the market
  - the rule earns only if the spread keeps returning to 0, that is, if the cointegration survives the formation period

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A1: an Engle–Granger statistic [Solved]

Task

- Two $I(1)$ series, $T = 200$. Step 1 gives residuals $\hat{u}_t$; step 2 gives $\Delta\hat{u}_t = -0.118\,\hat{u}_{t-1}$, with SE $= 0.036$.
- 1. Compute τ.
- 2. Compare τ with the 5% and 10% Engle–Granger values for 2 variables, -3.37 and -3.07, and decide.
- 3. Say what the Dickey–Fuller value -2.88 would have concluded.
- 4. Explain why the two critical values differ.
- Report: one number, two decisions and one sentence.

Solution

- 1. $\tau = -0.118/0.036 = -3.28$
- 2. $-3.28 > -3.37$: do not reject "no cointegration" at 5% ($p = 0.058$); $-3.28 < -3.07$: reject at 10%
- 3. $-3.28 < -2.88$: the Dickey–Fuller table would wrongly report cointegration at 5%
- 4. OLS chooses $\hat{b}$ to minimise the residual variance, so $\hat{u}_t$ looks more stationary than a true error: the null distribution shifts to the left.

Back Exercise list

A2: three variables [Proposed]

Solution discussed in class

Task

- Three $I(1)$ series, $T = 150$: y_t regressed on x_{1t} and x_{2t}. Step 2 gives $\hat{\gamma} = -0.162$, SE $= 0.041$. Model: A1.
- 1. Compute τ.
- 2. Decide at 5% and at 1% with the values for 3 variables, -3.80 and -4.39.
- 3. Say which critical value a student would wrongly use if he forgot x_{2t} (2 variables: -3.38).
- 4. Say how many cointegrating vectors the Engle–Granger method can find with three variables.
- Report: one number, two decisions and two sentences.

Back Exercise list

A3: an error correction model [Solved]

Task

- Estimated ECM (log consumption c_t, log income y_t, in %): $\Delta c_t = 0.2 + 0.5\,\Delta y_t - 0.25\,(c_{t-\mathbf{1}} - 0.9\,y_{t-\mathbf{1}})$.
- 1. Give the long-run and the short-run effect of income on consumption.
- 2. Interpret the coefficient -0.25 and compute the half-life.
- 3. Compute Δc_t if $c_{t-\mathbf{1}} - 0.9\,y_{t-\mathbf{1}} = 2$ and $\Delta y_t = 1$.
- 4. Say what a coefficient of $+0.25$ would mean.
- Report: two elasticities, one half-life, one forecast and one sentence.

Solution

- 1. Long run: 0.9 (from the equilibrium $c = 0.9y +$ const.); short run: 0.5 (same period)
- 2. A quarter of last period's gap is closed each period; $h_{\mathbf{1/2}} = \ln 0.5 / \ln 0.75 = 2.41$ periods
- 3. $\Delta c_t = 0.2 + 0.5 \cdot 1 - 0.25 \cdot 2 = 0.2$: the error correction cancels the short-run effect
- 4. Consumption would move away from equilibrium after a gap: no error correction, no cointegration.

Back Exercise list

A4: a bivariate VECM [Proposed]

Solution discussed in class

Task

- $\Delta y_{1t} = -0.20\, z_{t-1} + u_{1t}$, $\Delta y_{2t} = 0.05\, z_{t-1} + u_{2t}$, with $z_t = y_{1t} - y_{2t}$. Model: A3.
- 1. Write α, β and $\Pi = \alpha\beta^\top$, and give the rank of Π.
- 2. Explain which variable does most of the adjusting.
- 3. Show that $z_t = (1 + \beta^\top \alpha)\, z_{t-1} + (u_{1t} - u_{2t})$ and compute its half-life.
- Report: one matrix, one rank, one half-life and one sentence.

A5: Johansen statistics from eigenvalues [Solved]

Task

- Three $I(1)$ series, $T = 200$, case with a constant; the Johansen eigenvalues are 0.120, 0.045 and 0.008.
- 1. Compute the trace statistics for $r = 0$, $r \leq 1$ and $r \leq 2$.
- 2. Compute the maximum-eigenvalue statistics.
- 3. Compare with the 5% values (trace 29.80, 15.49, 3.84; maximum 21.13, 14.26, 3.84) and choose the rank.
- 4. Give the number of common trends.
- Report: six statistics, one rank and one number of trends.

Solution

- 1. $-200[\ln 0.880 + \ln 0.955 + \ln 0.992] = 36.38$; $-200[\ln 0.955 + \ln 0.992] = 10.82$; $-200 \ln 0.992 = 1.61$
- 2. 25.57; 9.21; 1.61
- 3. $r = 0$ rejected by both ($36.38 > 29.80$, $25.57 > 21.13$); $r \leq 1$ not rejected: rank 1
- 4. $n - r = 3 - 1 = 2$ common stochastic trends.

A6: from a VAR(2) to a VECM [Proposed]

Solution discussed in class

Task

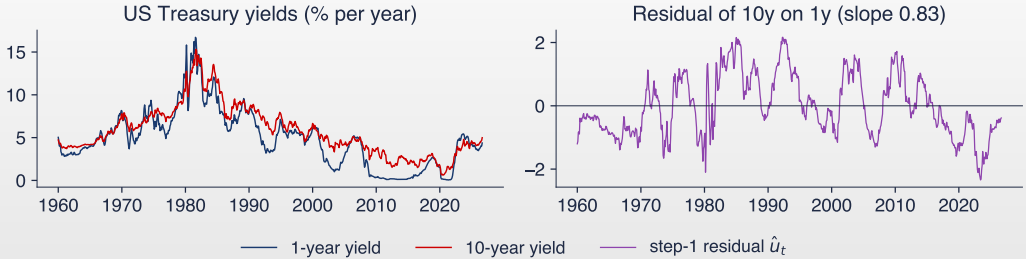
- $\mathbf{y}_t = A_1 \mathbf{y}_{t-1} + A_2 \mathbf{y}_{t-2} + \mathbf{u}_t$ with $A_1 = \begin{pmatrix} 0.7 & 0.2 \\ 0.1 & 0.7 \end{pmatrix}$, $A_2 = \begin{pmatrix} 0.1 & 0 \\ 0 & 0.2 \end{pmatrix}$. Model: A5.
- 1. Compute $\Pi = A_1 + A_2 - I$ and $\Gamma_1 = -A_2$.
- 2. Show that Π has rank 1 and write it as $\alpha\beta^\top$ with $\beta = (1, -1)^\top$.
- 3. Interpret the signs of α.
- Report: two matrices, α and one sentence.

B1: the 1-year and the 10-year US yields [Solved]

- **Question**: are the 1-year and the 10-year Treasury yields cointegrated, how fast does their gap close, and which rate closes it?
- **Inputs**: FRED GS1 and GS10, monthly averages, % per year, January 1960–September 2026
- **Tasks**
  1. Run ADF (with a constant) on both yields and on the change of the 10-year yield.
  2. Run Engle–Granger (10-year on 1-year) and Phillips–Ouliaris, and compare with the 5% Engle–Granger value.
  3. Estimate the ECM of the 10-year yield (one lag of each difference) and the same equation for the 1-year yield.
  4. Interpretation: if the 10-year yield is 1 percentage point above its equilibrium, how long until half of the gap is closed, and by which rate?
- **Report**: three ADF p-values, two cointegration tests, two adjustment coefficients and one sentence
- Notebook: section B1

Back Exercise list

B1: solution [Solved]



- ADF p = 0.26 (1 year), 0.63 (10 years): unit roots; the change: p < 0.001; both yields $I(1)$
- Step 1: $\hat{\imath}_t^{(10)} = 1.76 + 0.827\, i_t^{(1)}$; step 2: $\tau = -3.59$ (19 lags) < -3.34: cointegration at 5% (p = 0.025); Phillips–Ouliaris $Z_t = -3.65$ (p = 0.022)
- ECM, 10 years: $\hat{\gamma} = -0.0295$ ($t = -4.81$), $\hat{\delta}_0 = 0.53$; 1 year: $\hat{\gamma} = 0.0172$ ($t = 1.25$), not significant
- Interpretation: half-life $\ln 0.5 / \ln(1 - 0.0295) = 23.1$ months, about two years; the 10-year yield closes the gap, the 1-year yield (set by monetary policy) does not react

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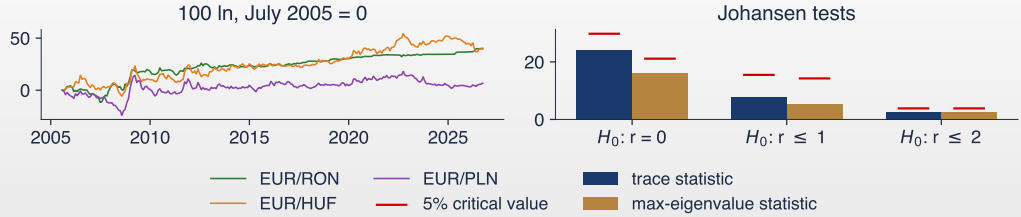
B2: consumption and GDP in Romania [Proposed]

- **Question**: is Romanian household consumption cointegrated with GDP, as the "great ratios" suggest?
- **Inputs**: Eurostat, real household consumption and real GDP, chain-linked volumes, seasonally adjusted, 1995–2026; c_t, y_t as $100 \ln$; model: B1
- **Tasks**
  1. Run ADF (constant and trend) on c_t and y_t, and ADF (constant) on their changes.
  2. Run Engle–Granger (c_t on y_t) and Phillips–Ouliaris, and report the slope.
  3. Plot the ratio of consumption to GDP and run ADF on $c_t - y_t$.
  4. Interpretation: does a stable consumption share make sense for Romania over 1995–2026?
- **Report**: four ADF tests, two cointegration tests, one chart and two sentences
- Notebook: section B2

B3: three Central European currencies [Solved]

- **Question**: do EUR/RON, EUR/HUF and EUR/PLN share a long-run equilibrium?
- **Inputs**: BNR reference rates (cross rates for HUF and PLN), month-end, 100 ln, since July 2005
- **Tasks**
  1. Run ADF on each rate.
  2. Run the Johansen trace and maximum-eigenvalue tests with a constant and lags by BIC.
  3. Compare with the three pairwise Engle–Granger tests and with the correlations of levels and of changes.
  4. Interpretation: should the three rates be modelled with a VECM?
- **Report**: three ADF p-values, a table of six Johansen statistics, a rank and one sentence
- Notebook: section B3

B3: solution [Solved]



	$r = 0$	$r \leq 1$	$r \leq 2$
trace (5% value)	24.21 (29.80)	7.94 (15.49)	2.72 (3.84)
maximum eigenvalue (5% value)	16.27 (21.13)	5.21 (14.26)	2.72 (3.84)

- ADF p = 0.55; 0.75; 0.08: three $I(1)$ rates; Johansen ($T = 255$, $k = 1$): no statistic exceeds its critical value: rank 0
- Engle–Granger p = 0.29 (RON, HUF), 0.08 (RON, PLN), 0.20 (HUF, PLN); correlation 0.89 in levels but only 0.31 in monthly changes
- Interpretation: no; with rank 0 there is no β to estimate; model the changes with a VAR in differences (Chapter 6)

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B4: ROBOR, the Romanian 10-year yield and Euribor [Proposed]

- **Question**: how many long-run relations tie Romanian interest rates to each other and to Euribor, and which rate adjusts?
- **Inputs**: ROBOR 3M and Euribor 3M (Eurostat), Romanian 10-year yield (monthly mean of daily data), January 2010–August 2026; model: B3
- **Tasks**
  1. Run the Johansen tests (constant, BIC lags) and choose the rank.
  2. Estimate the VECM with that rank and the constant restricted to the cointegrating relation; normalise β on ROBOR.
  3. Report $\hat{\alpha}$ with t-statistics and run ADF on the equilibrium error.
  4. Interpretation: which rates are weakly exogenous, and what does this say about Romanian monetary conditions?
- **Report**: a table of Johansen statistics, $\hat{\beta}$, $\hat{\alpha}$ with t, one ADF test and two sentences
- Notebook: section B4

C1: pairs trading on two banks [Proposed]

- **Question**: would a trader who found Banca Transilvania and BRD cointegrated at the end of 2021 have earned money afterwards?
- **Inputs**: adjusted daily prices, formation 2014–2021, trading January 2022–September 2026; models: B1, B3
- **Tasks**
  1. On the formation period, run Engle–Granger and Phillips–Ouliaris (TLV on BRD) and keep $\hat{b}$, the mean and the standard deviation of the spread.
  2. On the trading period, apply the z-score rule (Prerequisites for Today (14/15)): open at $|z| > 2$, close at $z = 0$; compute the annual return and the Sharpe ratio, gross and net of 0.20% per unit traded.
  3. Repeat the Engle–Granger test on the trading period alone.
  4. Interpretation: is the result evidence that pairs trading works on the BVB?
- **Report**: two tests, the number of trades, two returns, two Sharpe ratios and a plan for a project
- Notebook: section C1

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C2: audit an AI answer [Proposed]

- **A student asked an AI assistant about cointegration. The answer:**
- `(a) The Engle-Granger statistic is -3.0 with two variables; since -3.0 < -2.87, the 5% Dickey-Fuller value, the series are cointegrated.`
- `(b) The ECM coefficient of the lagged equilibrium error is +0.15, so 15% of the gap is corrected each period.`
- `(c) Johansen rejects r = 0 and r <= 1 but not r <= 2 for three yields, so the yields share two common trends.`
- `(d) EUR/RON and EUR/HUF are both I(1) and R2 = 0.79 in levels, so they are cointegrated.`
- `(e) If two series are cointegrated, at least one of them Granger-causes the other.`
- `(f) A pairs-trading rule with a Sharpe ratio of 0.44 on 2014-2026 data will keep this Sharpe ratio in the future.`
- **Tasks**
  1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
- **Report**: a list of six verdicts with one line of justification each

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Key takeaways

- Test that every series is $I(1)$ first; then test the equilibrium error, never the correlation of levels
- Residual-based tests need Engle–Granger critical values, which depend on the number of variables
- The ECM gives the speed of adjustment; the half-life $\ln 0.5 / \ln(1+\gamma)$ is in the units of the data
- Johansen gives the rank; $n - r$ is the number of common trends; α shows who adjusts
- An AI answer is a draft: check the critical values, the signs and the number of trends

After the seminar

- Details: Chapter 7
  - common trends, the Engle–Granger distribution, ECM, the VECM and the Granger representation theorem, the Johansen tests, forecasting and pairs trading
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: pairs trading on the BVB, out of sample and after costs
- Reading: Huang and Petukhina (2022), Ch. 9; Hamilton (1994), Ch. 19–20; Engle and Granger (1987)

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